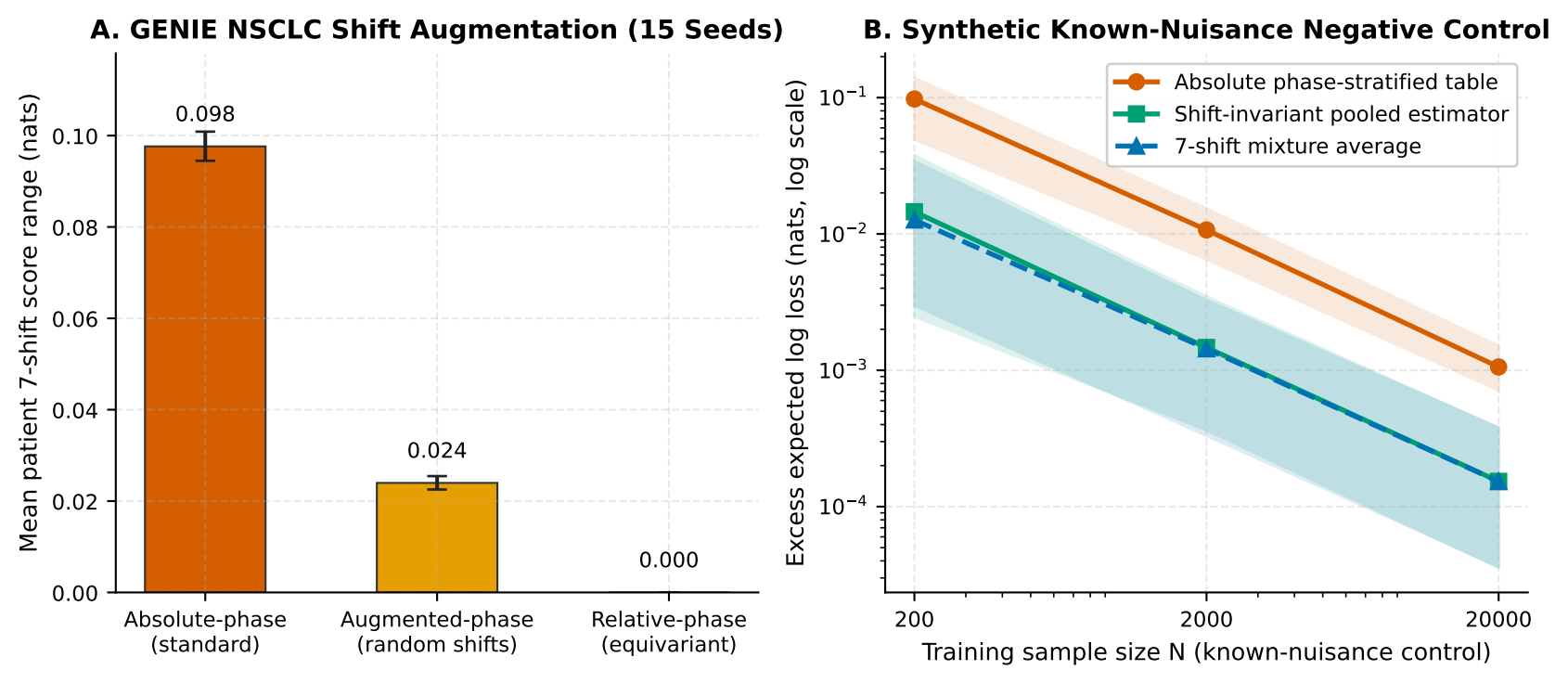
A. GENIE NSCLC Shift Augmentation (15 Seeds)
Mean patient 7-shift score range (nats)
0.00
0.02
0.04
0.06
0.08
0.10
0.098
0.024
0.000
Absolute-phase (standard)
Augmented-phase (random shifts)
Relative-phase (equivariant)
B. Synthetic Known-Nuisance Negative Control
Excess expected log loss (nats, log scale)
10^{-1}
10^{-2}
10^{-3}
10^{-4}
Absolute phase-stratified table
Shift-invariant pooled estimator
7-shift mixture average
200
2000
20000
Training sample size N (known-nuisance control)